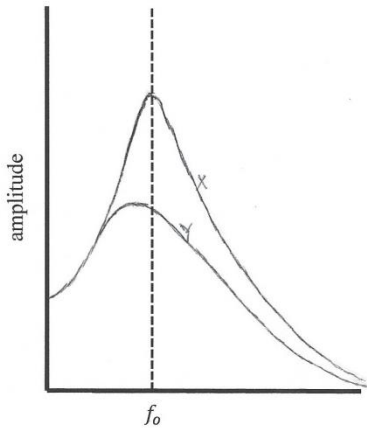


			Values on the axes of both graphs (1)		4		4	2	
		(v)	$mgh = 3.8 \times 10^{-3} \text{ J}$ ecf (1) $h = \frac{3.8 \times 10^{-3}}{0.05 \times 9.81} = 0.008 \text{ m}$ (1) (= 8 mm) Or: using trig: $4 - 4 \cos (0.25/4)$ (1) = 8 mm (1)		2		2	2	
	(b)	(i)	 amplitude f_0 frequency of driving force Sensible curve [X] with label [accept curve to the origin]	1			1		
		(ii)	Resonance	1			1		
		(iii)	Sensible curve with label (see part (b) (i) [cannot cross X – always below, can co-incide at low frequencies] (1) Lower maximum amplitude at the same frequency or to left of maximum of curve X (1)	2			2		
			Question 7 total	5	11	0	16	8	0